NIT ARUNACHAL PRADESH

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DUNE DOMINATOR

Conquer the Terrain. Command the Machine.



Introduction

Dune Dominator is a two-round off-road robotics challenge. Teams build a rugged, load-carrying rover that must cross a rough, obstacle-filled arena while keeping as much of its sand load as possible.

It is not a race of speed alone. It tests traction, balance, control and engineering judgement under pressure, and every bump, slope and boundary punishes weak design choices.

Event Description

Teams of two to four members design a rover that hauls sand across uneven, mechanically demanding terrain while losing as little material and time as possible. From wheel and suspension design to motor control and overall stability, every engineering decision can decide between getting stuck and conquering the dunes.

The event will be conducted physically on the Institute campus.

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Event Format

ROUND 1 — ELIMINATION ROUND

- Each team consists of 2 to 4 members (team leader included).
- The rover carries 500 g of sand inside the sand carrier, with the lid properly fixed.
- The rover must cross the complete arena and clear every obstacle on the way.
- Standard gameplay and technical rules apply. Performance in this round decides qualification for Round 2.

ROUND 2 — FINAL ROUND

- Only the teams qualified from Round 1 take part.
- 750 g of sand is placed in the sand carrier, with the top left open (no lid).
- The rover must cross the entire arena while losing as little sand and taking as little time as possible.
- Extra obstacles or changes to the arena may be introduced and revealed on the spot.
- The final rankings are decided on the final score of this round.

REGISTRATION FEE

- ₹50 per member.

Arena Specification

- The arena layout, obstacle set-up and detailed specifications will be revealed on the event day.
- Round 1 and Round 2 may use different arenas, with different layouts, obstacle types and difficulty.
- The arena imitates uneven off-road terrain and tests the rover's stability, traction and load retention.
- It includes tunnels, bridges and banked sections, along with surprise obstacles revealed during the event.
- The terrain is uneven and mechanically demanding, so build a robust, stable and adaptable rover.

Rover Specification

- Only self-powered rovers are allowed, with a maximum battery voltage of 24 V.
- The rover must be controlled manually at all times through a wireless control system. Autonomous or pre-programmed operation is strictly prohibited.
- Maximum rover weight: 2 kg (excluding the sand).
- Maximum dimensions: 40 cm x 30 cm x 30 cm (L x B x H).
- Motors: there is no speed restriction.
- Structural material: PVC or plywood is permitted.

SAND CARRIER

- A cuboidal box of 10 cm x 10 cm x 5 cm (L x B x H).
- Made of wood sheet with a minimum thickness of 3 mm.
- Mounted on the upper side of the rover.
- A separate, removable lid must be provided to cover the top of the carrier.

WHEELS AND TRACKS

- The rover must use wheels or continuous tracks to move. There is no limit on their number.
- Spikes, claws, screws, sharp protrusions or any feature that can damage the arena are prohibited.
- Active traction mechanisms that change the wheel or track geometry during a run are not permitted.
- Wheels or tracks must be non-metallic.

SUSPENSION

- Passive suspension systems are permitted.
- Active or electronically controlled suspension is not allowed.
- Suspension parts must be mechanically stable and must not change the rover's geometry during a run.

Gameplay Rules

- The rover must start from the designated start point when the timer begins.
- The rover must cover the complete track and cross all the obstacles.
- A maximum of 5 hand touches are allowed if the rover gets stuck or goes off track.
- Penalty points are deducted if the rover enters the designated Penalty Zone or touches the arena boundaries.
- The rover should keep sand loss and completion time to a minimum.

Judging Criteria

- Sand Retention: the amount of sand left in the carrier at the end of the run.
- Time Efficiency: the time taken to reach the designated end point.
- Stability and Control: stability of the rover on the course, including body roll, pitching and control on uneven terrain.
- Operational Discipline: following the gameplay rules, including hand touches, boundary violations and entry into Penalty Zones.
- The detailed scoring method, the weightage of each factor and the penalty values will be revealed on the event day and applied equally to all teams.

Fair-Play and Rover Rules

1. Inter-college teams are permitted.
2. The rover must not intentionally damage the arena or other teams' rovers.
3. Once the competition starts, no changes, disassembly or modifications to the rover are allowed until the results are declared.
4. Any form of cheating, tampering with the arena or intentional interference leads to disqualification.
5. The organizers may modify the rules, arena specifications or judging criteria for a smooth and fair competition, and any change will be announced.
6. The decisions of the judges and event managers are final and binding in case of any dispute.

Final Ranking

Round 1 determines qualification by completing the arena. Round 2 determines the final ranking using the final score of that round.

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*PRIZEPOOL

Prizes worth ₹20,000 to be won.

CONTACTS

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